MILITARY SPECIFICATION
DISCRIMINATOR, DISCRETE SIGNAL MD-736( )/A

1. SCOPE

1.1 This specification covers one type of discriminator, which provides for rejection and acceptance of various signal levels, designated Discriminator, Discrete Signal MD-736( )/A. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-20	Bag, Cotton, Mailing
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
NN-P-71	Pallets, Material Handling, Wood, Double Faced, Stringer Construction
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure Sensitive Adhesive
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed A and Lock-Corner
PPP-B-636	Box, Fiberboard
QQ-S-781	Steel Strapping, Flat

MILITARY

MIL-T-27	Transformer and Indicator, General Specification For
MIL-P-116	Preservation, Methods Of
MIL-B-4301	Box, Water Resistant Paperboard, Folding, Set-up and Metal-Stayed
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

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SPECIFICATIONS

MILITARY

MIL-S-19500/256	Semiconductor Device, Diode, Silicon, Switching Types 1N643, 1N662, 1N663
MIL-S-19500/286	Semiconductor Device, Diode, Silicon, Power Rectifier Types 1N4245 Through 1N4249
MIL-I-46058	Insulation Compound, Electrical
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads; (40 in X 48 in 4-Way Partial and 4-Way Pallets)
MIL-STD-252	Wired Equipment, Classification Of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements For Electronic Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

USAECOM

ES-DL-201057	Discriminator, Discrete Signal MD-736( )/A
SC-A-678019	Combined Burn-In/Reliability Requirements For Discriminator, Discrete Signal MD-736( )/A

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

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3.1 Description. - Discriminator, Discrete Signal MD-736( )/A is a separately housed unit containing a printed circuit board inclosed in a metal cover. The unit provides filtering of transmit audio signals from very-high frequency (vhf), ultra-high frequency (uhf), and high frequency (hf) radio signals that pass through interphone systems in aircraft. The unit will disable frequency-modulation (fm) reception when vhf, uhf, or hf systems are being used. Four (4) input channels are available and operation is from a power source of 28 volts dc. Wires pass through one side of the unit for connection to applicable systems. Holes in the base provide for mounting the unit to its applicable location.

3.2 Construction. - The units on order shall be constructed in accordance with the following drawing list:

ES-DL-201057	Discriminator, Discrete Signal MD-736( )/A

3.3 Cleaning. -

3.3.1 Parts. - Prior to the application of conformal coating, on parts affected, and after fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Post cleaning assembly processes shall minimize handling by bare hands to prevent contamination by fungus nutrients. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, and other parts.

3.3.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4 Finish, protective. - The units shall be given protective finish in accordance with Specification MIL-F-14072, and the applicable drawings. (See 4.4)

3.5 Securing of parts. - Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6 Tropicalization of materials. - Materials shall be treated as follows:

3.6.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

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3.6.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.6.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.6.4 Statement of treatment. - The contractor shall submit for approval, to the Contracting Officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

3.7 Marking (See 4.4). -

3.7.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be engraved in accordance with Group 1 of the specification.

3.7.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the unit(s).

3.7.3 Serial numbers. - The following unit(s) shall be serial numbered:

Discriminator, Discrete Signal MD-736( )/A

3.8 Connections, general. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.9 Conformal coatings (See 4.4). - Except as otherwise specified in the equipment drawings, printed wiring assemblies shall be conformally coated with a coating material which conforms to MIL-I-46058, Type PUR (polyurethane). The coating shall be applied to both sides of the cleaned printed wiring assembly including the component leads. These assemblies shall be cleaned of flux and other contaminants prior to application of coating. Cleaning compounds shall have no deleterious effects on any part of the printed wiring assembly. The conformal coating shall be compatible with all parts of the printed wiring assembly and the thickness shall be 0.0035 $\pm$ 0.0005 inch nominal tolerance. The coated assemblies shall show no

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visible bubbling, blistering, wrinkling, cracking, pinholes, or peeling of the coating material or corrosion of printed conductors. Printed wiring assemblies shall be adequately masked or otherwise protected in such a manner that application of conformal coating does not degrade the electrical performance of the assembly. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Electrical and mechanical mating surfaces such as connector contact points, test points, screw threads, bearing surfaces, etc., shall not be coated. When masking those areas which are not to be coated, and tape is used, the tape shall be an electrical grade. When the printed wiring assembly utilizes glass encased components, such components shall be protected against breakage by the conformal coating by covering the component with thin pliant buffer material, prior to coating, such as polyvinylidene fluoride or silicone rubber. This buffer material shall be compatible with the conformal coating material and all parts of the printed wiring assembly. In addition, the buffer material shall be fungus and flame resistant; and shall be clear transparent so markings on the component shall be visible. Conformal coatings shall be clear transparent so markings on the components shall be visible.

3.10 Semiconductor (Diodes) (See 4.4). - These electron devices (diodes) shall conform to MIL-S-19500 and Detailed Specification Sheets 256 and 286.

3.11 Transformers (See 4.4). - The audio transformers shall conform to MIL-T-27, Type TF5RX13ZZ. Audio frequency 300 Hz - 10 kHz, maximum mv 175 v peak; primary impedance 4K ohms, 2.5 ma DC unbalanced; secondary impedance 600 ohm split. The DC primary resistance shall be between 8 and 14 ohms.

3.12 Soldering. -

3.12.1 Soldering. - Soldering shall conform to Requirement 5 of MIL-STD-454 and the equipment drawings.

3.12.2 Rosin-base residues. - Flux residues shall be removed by applying organic solvents. Mechanical means such as agitation, brushing, etc., may be used in conjunction with the solvents. The cleaning solvents and methods used shall not have any deleterious effect on the parts, materials, and assemblies being cleaned.

3.12.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of

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loose particles of flux, brush bristles, or other foreign material remaining in the unit(s), flux being spread over a larger area, or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.13 Service conditions (See 4.7). - The unit(s) shall meet the service condition as specified in Specification MIL-E-5400 for Class IA equipment and for application in both fixed wing aircraft and helicopter, and the specification requirements, except as otherwise specified herein.

3.13.1 Temperature-Altitude combination (See 4.7.1). - The unit(s) shall meet the temperature-altitude requirements of MIL-E-5400 for Class IA equipment except the lower non-operating limit shall be -54°C.

3.13.2 Air transport (See 4.7.2). - The unit(s) shall withstand air shipment at altitudes to 50,000 feet above sea level.

3.13.3 Relative humidity (See 4.7.3). - Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. This test shall be performed over a 10 day cycle. The unit(s) shall withstand these conditions during continuous operations, intermittant operations, short term operations or exposure in a non-operating condition.

3.13.4 Sand and dust (See 4.7.4). - The unit(s) shall withstand exposure to sand and dust particles as encountered in desert areas.

3.13.5 Vibration (See 4.7.5). - The unit(s) shall meet the vibration Curve IV for combination aircraft and helicopter equipment.

3.13.6 Shock (See 4.7.6). -

3.13.6.1 Benchhandling (See 4.7.6.1). - The unit(s) shall be capable of full specification performance after subjection to the required number of drops.

3.13.6.2 Equipment (See 4.7.6.2). - The unit(s) shall meet the equipment shock test. The unit(s) shall be capable of meeting full specification performance after the shock test.

3.13.7 Fungus (See 4.7.7). - The unit(s) shall show no evidence of viable fungus after being subject to 28 days of fungus test.

3.13.8 Salt fog (See 4.7.8). - The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

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3.13.9 Acceleration (See 4.7.9). - The equipment shall be operable and sustain no physical damage when exposed to acceleration levels of up to 6 g's.

3.13.10 Explosive proof (See 4.7.10). - The unit(s) shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.14 Electrical Requirements. -

3.14.1 Large signal insertion loss (See 4.8.1). - With an input level of 0.5 volt RMS at 1.0 kHz from a 150 ohm source, the insertion loss of each audio circuit shall not be greater than 6 Db. With the level raised to 2.7 volts RMS, the insertion loss of the circuit shall be 2.5 Db $\pm$ 1 Db.

3.14.2 Small signal rejection (See 4.8.2). - With an input level of 20 millivolts RMS at 1.0 kHz from a 150 ohm source, each audio circuit loss shall be no less than 55 Db.

3.14.3 Adjacent channel isolation (See 4.8.3). - With an input level of 2.5 volts RMS at 1 kHz from a 150 ohm source, adjacent circuit isolation between any combination of the three audio circuits shall be less than -40 dbm below the generator reading in dbm.

3.14.4 FM circuit isolation (See 4.8.4). - With in input voltage of 2.5 VRMS at 1.0 kHz from a 150 ohm source impedance across Terminal 4 of Relay K1 and Terminal 19 (with Relay K1 open) the isolation from audio circuits shall be equal to or greater than 100 Db.

3.14.5 Audio circuit isolation (See 4.8.5). - The audio common circuits shall be isolated from the equipment case and bypassed for RF.

3.14.6 Input power (See 4.8.6). - The equipment shall operate directly from a nominal 27.5 DC aircraft power as defined by MIL-STD-704, Category B. Cutoff types of transient suppressors shall not be used.

3.15 Combined Burn-In/Reliability (See 4.6). - Each unit on order shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the Combined Burn-In/Reliability as specified in SC-A-678019. The MTBF of the equipment shall be 3000 hours.

3.16 Operational inspection (See 4.9). - Each unit(s) shall be subjected to the operation inspection as specified, after applicable processes, inspection and tests are completed, to determine that each unit is operable prior to packaging.

3.17 Interchangeability (See 4.10). - Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable without modification of such items or the equipment.

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3.18 Workmanship. - The unit(s) shall be manufactured and assembled in accordance with the drawings (See 3.2) and applicable portions of the following requirements and paragraphs herein:

MIL-STD-454:

Soldering	Requirement 5
Workmanship	Requirement 9
Fastener hardware	Requirement 12
Structural Welding	Requirement 13

Paragraphs herein:

Cleaning	3.3
Securing of parts	3.5
Tropicalization	3.6
Connections	3.8

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12 and 4.13)

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4.3 First Article inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (See Tables I and II and III respectively). The first article inspection will normally be performed in this order:

SAMPLE 1: (1) Temperature/Altitude; (2) Humidity: (3) Vibration; and (4) Sand and Dust.

SAMPLE 2: (1) Vibration; (2) Temperature/Altitude; (3) Humidity: (4) Shock, Bench Handling and Equipment; and (5) Salt Fog.

SAMPLE 3: (1) Explosion Proof; (2) Acceleration; (3) Fungus, other first article inspections may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

ITEM	WHERE REQUIRED
Finish Protective	3.4
Marking	3.7
Conformal Coating	3.9
Semiconductors	3.10
Transformers	3.11

4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to Combined Burn-In/Reliability (See 3.15) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of Burn-In (debugging) shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - - at its discretion - - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105, using the general inspection levels and the AWL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection 4.9 shall be last.

TABLE I - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.18	4.11		
Printed Circuit Board			0.65%	2.5%
MD-736( )/ARC (Cover off)			1.00%	4.0%
MD-736( )/ARC (Assembled)			1.00%	4.0%
Electrical:				
Large Signal Insertion Loss	3.14.1	4.8.1	*1.0% for the Combined Group	
Small Signal Rejection	3.14.2	4.8.2		
Adjacent channel isolation	3.14.3	4.8.3		
FM Circuit Isolation	3.14.4	4.8.4		
Operational Inspection	3.16	4.9	0.1%	

*All electrical defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105, using the special insertion levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group inspection.

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4.5.3 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II - Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Audio Circuit Isolation	3.14.5	4.8.5	6.5%
Input Power	3.14.6	4.8.6	Combined Group
Interchangeability	3.17	4.10	6.5%

4.5.4 Group C inspection. - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

TABLE III - Group C Inspection

Inspection	Req. Para.	Insp. Para.
Subgroup I:		
Temperature-Altitude	3.13.1	4.7.1
Relative Humidity	3.13.3	4.7.3
Vibration	3.13.5	4.7.5
Subgroup II:		
Sand and Dust	3.13.4	4.7.4
Bench handling	3.13.6.1	4.7.6.1
Equipment	3.13.6.2	4.7.6.2
Subgroup III:		
Fungus	3.13.7	4.7.7
Salt Fog	3.13.8	4.7.8
Acceleration	3.13.9	4.7.9
Air Transport	3.13.2	4.7.2
Explosive Proof	3.13.10	4.7.10

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4.5.4.1 Sampling for inspection. - Units shall be selected at random for each group of inspections in Table III as specified in 4.5.4.1.1 thru 4.5.4.1.4. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot(s). Selection of samples shall be as specified and within 45 calendar days after selection. The samples shall have been subjected to and completed the Group C inspections (See 6.3).

4.5.4.1.1 Subgroup I. - For this subgroup, two units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent inspection, two units per month shall be selected.

4.5.4.1.2 Subgroup II. - For this subgroup, two units (these units may be the units selected for above subgroup I), from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units every two (2) months shall be selected.

4.5.4.1.3 Subgroup III. - For this subgroup, two units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units shall be selected every four months except that fungus shall be selected when one-half of the units on order have been accepted by the Government QA Representative.

4.5.5 Non-compliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspection, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

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4.6 Combined Burn-In/Reliability. - Each unit on order shall be subjected to and meet the Burn-In/Reliability requirements as specified in SC-A-678019. Specification MIL-STD-781 shall apply as specified in this drawing.

4.7 Service conditions test. - The unit(s) shall be tested in accordance with Specification MIL-STD-810, except as otherwise specified. Electrical tests shall be performed before and after all service condition tests to verify compliance with the applicable paragraphs 4.8.1, 4.8.2, 4.8.3, except as otherwise specified. No defects shall be allowed.

4.7.1 Temperature-altitude comb. - The unit(s) shall meet the test of Method 504, Procedure I of MIL-STD-810, except the lower non-operating temperature shall be -54°C, for compliance with 3.13.1. An electrical test per 4.8.1 shall be performed during the test, as required.

4.7.2 Air transport. - The unit(s) shall be subjected to a simulated altitude of 50,000 feet for four (4) hours for compliance with 3.13.2.

4.7.3 Relative humidity. - The unit(s) shall be subjected to and meet the test of Method 507, Procedure II of MIL-STD-810. In Step 6, measurements per 4.8.1 shall be taken during the last five (5) hours of each test cycle for compliance with 3.13.3.

4.7.4 Sand and dust. - The unit(s) shall meet the test of Method 510, Procedure I of MIL-STD-810, for compliance with 3.13.4.

4.7.5 Vibration. - The unit(s) shall be subjected to the test of Method 514, Procedure I, Curve Z (Parts 1, 2 and 3) of MIL-STD-810, for compliance with 3.13.5.

4.7.6 Shock. -

4.7.6.1 Benchhandling. - The unit(s) shall be assembled for servicing and placed in a suitable position for servicing on a solid two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other edges of the same horizontal face as axis for a total of four (4) drops.

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b. Repeat a. above, with the assembly resting on other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. The tests of this paragraph shall be performed to determine compliance with 3.13.6.1.

4.7.6.2 Equipment. - The unit(s) shall be subjected to the test of Method 516, Procedures I and III. After the Procedure I test, the unit(s) shall be tested as required in 4.8.1 and 4.8.2 to determine compliance with 3.13.6.2. No electrical test is required after the Procedure III test.

4.7.7 Fungus. - The unit(s) including accessories shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The unit(s) shall be subjected to the test in this open condition and as specified for compliance with 3.13.7 and shall not be required to be operated during or after their test.

4.7.8 Salt fog. - Unit(s) shall be exposed in normal operating condition to salt fog in accordance with Method 509 of MIL-STD-810, for 48 hours. The unit(s) shall not be opened and shall not be required to be operated. The unit(s) shall meet the requirement of 3.13.8.

4.7.9 Acceleration. - The unit(s) shall be tested in accordance with Method 513, 3.3.5 of MIL-STD-810. An electrical test, in accordance with 4.8.1, shall be conducted during the test. This test shall be made to determine compliance with 3.13.9.

4.7.10 Explosive proof. - Unit(s) shall be subjected to the test of Method 511, Procedure I of MIL-STD-810, for compliance with 3.13.10. An electrical test per 4.8.1 shall be performed during the test.

4.8 Electrical tests. -

4.8.1 Large signal insertion loss. - Connect unit under test as per Figure 1 (attached hereto). Adjust audio oscillator level (from 150 ohm source) to 0.5 volts RMS. Record input level in Db. Record input level of unit under test in Db. Calculate loss in Db. Repeat test with the level raised to 2.7 volts RMS. The requirements of 3.14.1 shall be met.

4.8.2 Small signal rejection. - Connect unit under test as per Figure 2 (attached hereto). Follow test procedure as noted. The requirements of 3.14.2 shall be met.

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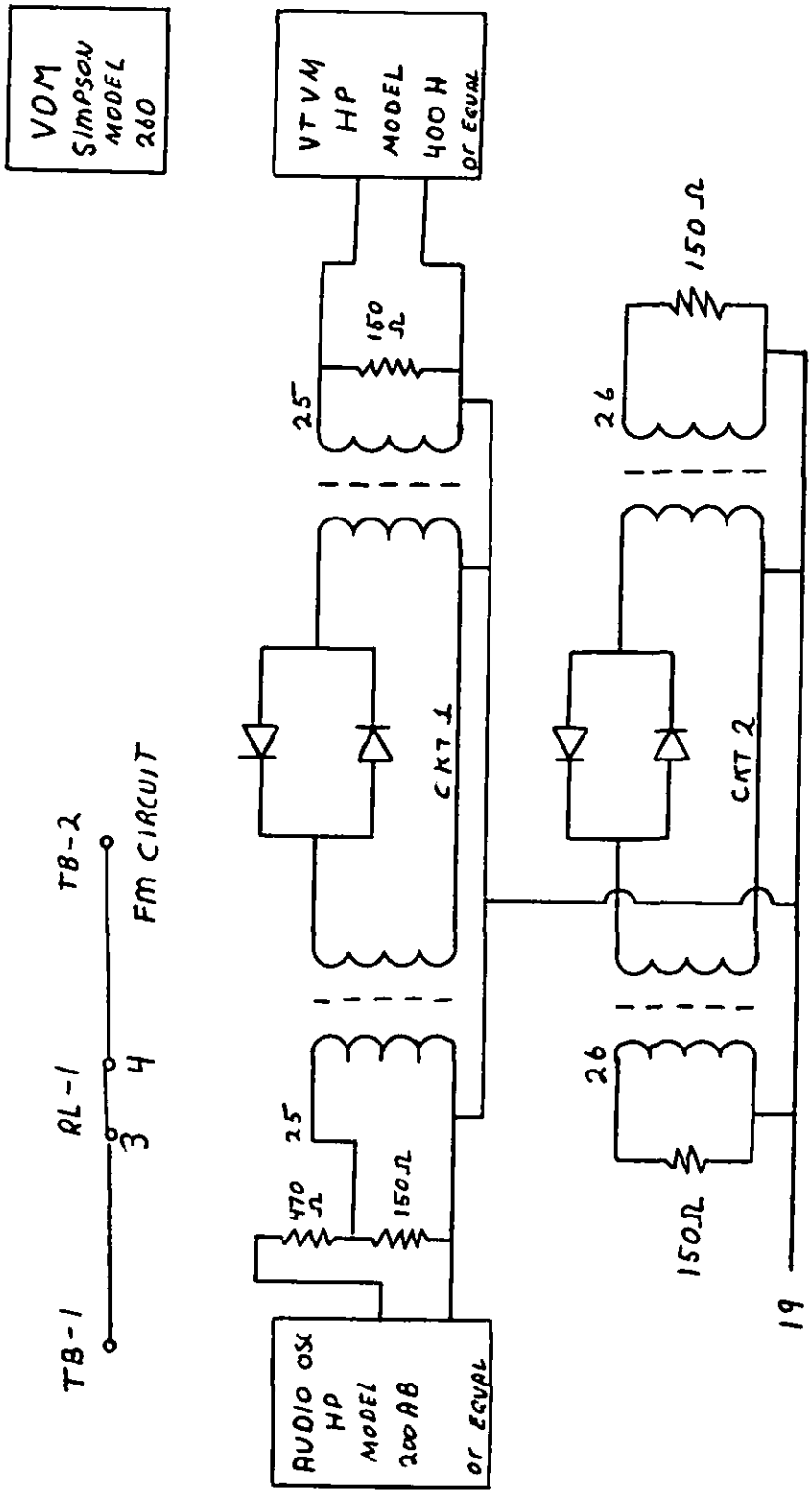


Figure 1 Dynamic Tests Schematic Diagram.

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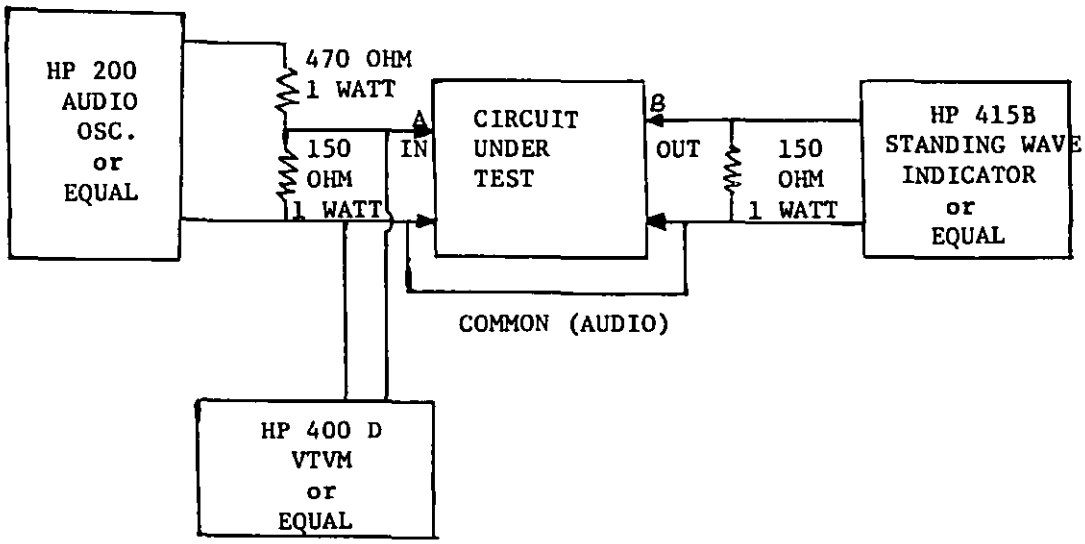


Fig. 2

Test Procedure:

1. Arrange test equipment as shown above.
2. Set audio oscillator frequency to 1 kHz.
3. Set audio oscillator level to provide a VTVM reading of 20 mv.
4. Remove lead "B" from the test circuit and connect it to lead "A".
5. Set "Range" control of Standing Wave Indicator to 30 db.
6. Set Input Selector of Standing Wave Indicator to 200 K _ _.
7. Adjust Gain control of Standing Wave Indicator to obtain a meter reading.
8. Adjust Frequency of audio oscillator for a peak reading on the Standing Wave Indicator.
9. Adjust Standing Wave Indicator gain control for a reading of zero db.
10. Remove lead B from lead A and again connect it to the circuit as shown.
11. Increase the range control until a meter reading is obtained.
12. The circuit attenuation is twice the value read on the Standing Wave Indicator.

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4.8.3 Adjacent channel isolation. - This test shall be conducted as follows:

a. Connect a 150 ohm resistor across the following terminals:

(1) #25 (TB-2) and #19 terminal

(2) #26 (TB-2) and #19 terminal

(3) #28 (TB-2) and #19 terminal

(4) #26 (TB-1) and #19 terminal

(5) #28 (TB-1) and #19 terminal

b. Connect an Audio Signal Generator and VTVM across #25 (TB-1) and #19 terminal.

c. Adjust output level of generator (set to 1.0 kHz) to 2.5 volts, record dbm. Measure output levels at terminals noted in 4.8.3.a (2), (3), (4) and (5). Record levels in dbm.

d. Remove 150 ohms load from #26 (TB-1) and #19. Connect a Signal Generator and VTVM (removed from #25 (TB-1) and #19) across #26 (TB-1) and #19. Connect a 150 ohm resistor across #25 (TB-1) and #19. Repeat test procedures as stated in 4.8.3.a (1), (3), (5), across #25 (TB-1) and #19 and 4.8.3.b and 4.8.3.c.

e. Remove 150 ohm load from #28 (TB-1) and #19. Connect a Signal Generator and VTVM (removed from #26 (TB-1) and #19) across #28 (TB-1) and #19. Connect a 150 ohm resistor across #26 (TB-1) and #19. Repeat test procedures as noted on 4.8.3a (1), (2), (4), across #25 (TB-1) and #19 and 4.8.3b and 4.8.3c. The requirements of 3.14.3 shall be met.

4.8.4 FM circuit isolation. - Connect signal generator with applicable termination to terminal #4 (Relay #1) and terminal #19. Connect 150 ohm loads to input and output transformer circuits. Set generator output level to 2.5 volts RMS. Energize relay with a 27.5 volt source. Read output level across primary and secondary windings of the audio transformers. Output levels shall be determined for compliance with 3.14.4.

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4.8.5 Audio circuit isolation. - With the unit(s) de-energized, tests shall be made between terminal #19 and the MD-736( )/A case for compliance with 3.14.5.

4.8.6 Input power. - Connect +27.5 volts to terminal #1 of Relay K1. Connect one end of a 1.5 volt light bulb to terminal #3 (Relay #1) the other end of the light bulb to the positive terminal of a 1.5 volt battery. Connect the negative terminal of the battery to terminal #4 (Relay #1). The bulb should light. Apply -27.5 volts to terminal #34 of terminal board #1. The Relay shall energize and the light bulb shall extinguish. With -27.5 volts applied to terminal #35 of terminal board #1, and then terminal #1 of terminal board #1, the relay shall energize and the light bulb shall extinguish when the voltage is applied to each terminal. These tests shall be conducted for compliance with 3.14.6.

4.9 Operational inspection. - The unit shall be connected under simulated operating conditions as follows: With 24 volts applied to Relay K1 and when each pin (#34, #35 and #1 of terminal board #1) is keyed, contacts #3 and #4 of Relay K1 shall open. Apply an audio signal to pins #25, #26 and #28 of terminal board #1, the signal at pins #25, #26 and #28 of terminal board #2 shall indicate a lower signal level. These tests shall be conducted for compliance with 3.16.

4.10 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance to the physical interchangeability requirements of 3.17. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides and mounting which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

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4.11 Visual and mechanical inspection. - Assembled unit(s) shall be examined for defects listed in MIL-STD-252. Prior to assembly into the unit(s), printed circuit boards shall be examined for compliance with MIL-P-55110 and the equipment drawings. Assemblies shall be examined for compliance with the equipment drawings and MIL-STD-252. The requirements of 3.18 shall be met.

4.12 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance with the packaging requirements of Section 5.

4.13 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract the operational inspection test 4.9, shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C, as specified. (See 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. - Discriminator, Discrete Signal MD-736( )/A shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Discriminator, Discrete Signal MD-736( )/A shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged per Method 1C-1.

5.1.1.4.2 Discriminator, Discrete Signal MD-736( )/A. - Each discriminator shall be individually packaged per Method III as follows: Cables and mounting hardware shall be placed within a cloth bag conforming to PPP-B-20, close bag with drawstrings. Cushion the unit together with the cloth bag by wrapping in paperboard conforming to PPP-T-45, Type III, Grade A. Place the cushioned item within a close-fitting box conforming to MIL-B-4301, Form 1, Style 11, Type A, Class A. Place the technical literature on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level C. - Discriminator, Discrete Signal MD-736( )/A, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be Level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2)

5.2.1 Level A. -

5.2.1.1 Consolidation. - A quantity of Discriminator, Discrete Signal MD-736( )/A packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, Class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers packed as specified in 5.2.1.1, shall be placed on a pallet, load Type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, Type IV, Size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, Class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, Overseas Type, PPP-B-621, Style 4, Class 2, or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 X 24 inches or more and the weight exceeds 100 pounds, 3 X 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specifications, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto, except that metal strapping shall conform to QQ-S-781, Type 1, Class B.

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5.2.2 Level B. -

5.2.2.1 Consolidation. - A quantity of Discriminator, Discrete Signal MD-736( )/A, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1 shall be pallitized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, Type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C. -

5.2.3.1 Consolidation. - A quantity of Discriminator, Discrete Signal MD-736( )/A, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be Class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be Class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping containers. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Discriminator, Discrete Signal MD-736( )/A is intended to be used in conjunction with Control, Intercommunication Sets, i.e., C-1611( )/A, and to reject low level signals while passing high level signals when used with airborne communication systems in both fixed wing and rotary wing type aircraft.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

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b. Levels of preservation, packaging, and packing required. (Level A, Level B, or Level C.)

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. First Article inspection:

(1) Several sample units of each item cited in Section 1 are generally required so that lengthy environmental and reliability tests can be completed on specified sample units while complete performance measurements can be made on the one sample unit. (See 3.1)

(2) Detailed requirements covering First Article pack(s), including inspection to be performed thereon and descriptive details, plans, drawings, etc.

(3) First Article pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Submission of the statement of treatment referenced in 3.6 should be as specified in contract. This statement should be submitted to the Contracting Officer.

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Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-55367(EL) DISCRIMINATOR, DISCRETE SIGNAL MD-736( )/A

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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